

TEST PLAN

Naretoi Charity Donation Management System

7.1. Introduction

7.1.1. Purpose

This test plan provides a comprehensive overview of the Naretoi Charity Donation Management System functionalities. The system is designed to facilitate online donations through multiple payment methods (M-Pesa and PayPal), manage donor information, track donations across four main support categories (Education Support, Medical Assistance, Building Support Facilities, and Food Aid), generate reports, and provide an intuitive user interface for donors and administrators.

Key System Functionalities:

- Online donation processing with M-Pesa and PayPal integration
- Donor registration and profile management
- Support category management (Education, Medical, Building, Food Aid)
- Real-time donation tracking and analytics
- Automated receipt generation and email notifications
- Admin dashboard for donation and donor management
- Reporting and data export capabilities

7.1.2. Testing Scope

This testing effort covers the following areas:

In Scope:

- User Interface (UI) testing for home page, donation forms, and navigation
- Donor registration and profile management functionality
- Donation processing workflow with payment gateway integration
- Support category display and selection mechanisms
- Admin panel functionality (Django Admin)
- Database operations (CRUD operations for donations, donors, categories)
- Email notification system for donation receipts
- Form validation and error handling
- Responsive design across devices (desktop, tablet, mobile)
- Security features (authentication, authorization, data protection)
- Payment gateway integration (M-Pesa and PayPal)

- Report generation and data export features

Out of Scope:

- Third-party payment gateway internal testing (M-Pesa/PayPal APIs)
- Load testing and performance benchmarking
- Penetration testing and advanced security audits
- Multi-language support (future enhancement)
- SMS notification system (future enhancement)
- Automated background tasks (Celery workers)

7.2. Testing Approach

7.2.1. Testing Objectives

The primary objectives of this testing effort are:

- **Identify and Report Bugs:**** Discover defects, errors, and inconsistencies in the system functionality
- **Ensure Feature Functionality:**** Verify that all implemented features work as per specifications
- **Validate User Experience:**** Ensure the application is user-friendly, intuitive, and accessible
- **Verify Data Integrity:**** Confirm accurate storage, retrieval, and processing of donation and donor data
- **Test Payment Integration:**** Validate seamless integration with M-Pesa and PayPal payment gateways
- **Assess Security:**** Verify authentication, authorization, and data protection mechanisms
- **Confirm Cross-Browser Compatibility:**** Test application across different browsers and devices
- **Validate Business Logic:**** Ensure donation calculations, category assignments, and reporting are accurate

7.2.2. Testing Levels

Strategy: Manual vs Automated Testing

Manual Testing:

- Exploratory testing of user interface and workflows
- User acceptance testing (UAT) with stakeholders
- Usability testing and accessibility checks
- Ad-hoc testing for edge cases
- Payment gateway integration testing (sandbox environments)

Automated Testing:

- Unit tests for models, views, and business logic
- Integration tests for module interactions
- Regression testing for future code changes

- API endpoint testing
- Database operation validation

Tools and Frameworks:

- **Unit Testing:** pytest, pytest-django
- **Code Coverage:** coverage.py
- **Test Data Generation:** model-bakery
- **Manual Testing:** Browser DevTools, Postman (API testing)
- **Version Control:** Git for tracking test artifacts

Types of Testing

1. Unit Testing

- **Purpose:** Test individual components in isolation
- **Scope:**
 - Model methods and properties (Donor, Donation, SupportCategory)
 - View functions and logic
 - Form validation
 - Utility functions
- **Tools:** pytest, Django TestCase
- **Coverage Target:** 80% code coverage minimum

2. Integration Testing

- **Purpose:** Test interactions between different modules
- **Scope:**
 - Donation workflow (donor creation → donation → payment)
 - Admin panel interactions with database
 - Email notification triggers
 - Payment gateway callbacks
 - Category-to-donation relationships
- **Tools:** pytest-django, Django Client
- **Focus Areas:** Database transactions, API endpoints, module communication

3. System Testing

- **Purpose:** Validate the complete integrated system
- **Scope:**
 - End-to-end donation process
 - Complete user journeys (home → category selection → donation → receipt)
 - Admin workflows (login → view donations → generate reports)
 - Cross-browser and responsive design testing
 - Security testing (authentication, CSRF protection)
- **Tools:** Manual testing, Browser Stack (optional)
- **Test Environments:** Development, Staging, Production sandbox

4. User Acceptance Testing (UAT)

- **Purpose:** Validate system meets user requirements and expectations

- **Scope:**
 - Real-world donor scenarios
 - Admin user workflows
 - Report generation and data export
 - Payment processing (with test accounts)
- **Participants:** Naretoi staff, selected donors (beta testers)
- **Duration:** 2 weeks
- **Acceptance Criteria:** 95% of test cases pass, no critical bugs

7.3. Test Cases and Results

7.3.1. Test Case Details

Module 1: Donor Management

Module 2: Support Categories

Module 3: Donation Processing

Module 4: User Interface

Module 5: Admin Panel

Module 6: Email Notifications

Module 7: Security

7.3.2. Status and Notes

	Test Steps	Expected Results
	1. Navigate to donation form 2. Enter valid donor details (name, email, phone, country) 3. Submit form	Donor record created in database v
st@example.com	1. Attempt to create donor with email test@example.com 2. Submit form	System rejects duplicate email with
	1. Enter invalid phone number (e.g., "12345") 2. Submit form	Form validation error displayed
	1. Access admin panel 2. Edit donor details 3. Save changes	Donor information updated success
	1. Select "Anonymous" option 2. Submit with minimal data	Donor created without personal det
	Test Steps	Expected Results
exist	1. Navigate to homepage 2. Verify category display	All 4 categories (Education, Medica
	1. Open donation form 2. Select category dropdown	All active categories available for s
tal	1. Complete a donation for category 2. Check category total_donations field	Total updated by donation amount
	1. Admin sets is_active=False 2. Check homepage and forms	Category no longer appears in pub
ge	1. Upload image via admin 2. Save category	Image stored and displayed correct
	Test Steps	Expected Results
ured	1. Fill donation form 2. Select M-Pesa 3. Enter amount and phone 4. Submit	Donation created with status "pend
ured	1. Fill donation form 2. Select PayPal 3. Enter amount 4. Submit	Donation created, redirected to Pay
	1. Enter amount < 100 KES 2. Submit	Validation error: "Minimum donation
	1. Enter USD amount 2. Submit donation	Amount converted to KES using cu
s	1. Simulate payment success callback 2. Update donation status	Status changes to "completed", tota
	1. Simulate payment failure 2. Check donation status	Status set to "failed", user notified
	1. Select anonymous option 2. Complete donation	Donor recorded as anonymous, no
	Test Steps	Expected Results
	1. Access homepage on desktop, tablet, mobile 2. Check layout and navigation	All elements responsive and acces
	1. Click all navigation links 2. Verify page routing	All links navigate to correct pages
	1. Submit empty form 2. Submit invalid data	Inline validation errors displayed fo
	1. Submit donation 2. Observe UI during processing	Loading spinner/message displaye
	1. Complete donation 2. View confirmation	Thank you message with donation
	Test Steps	Expected Results
	1. Navigate to /admin 2. Enter credentials 3. Login	Successful authentication, dashboa
	1. Login to admin 2. Navigate to Donations	List of all donations with filters disp
	1. Use status filter 2. Apply filter	Only donations matching status sho
	1. Select donations 2. Choose export format (CSV/Excel) 3. Export	File downloaded with correct data
	1. Edit donation 2. Change status to completed 3. Save	Status updated, related fields recal
	Test Steps	Expected Results
	1. Complete donation 2. Check donor email	Receipt email sent with donation de
	1. Review email content 2. Verify details	All details (amount, category, date)
	1. Trigger receipt send 2. Simulate email failure	Error logged, donation still marked
	Test Steps	Expected Results
	1. Submit form without CSRF token	Request rejected with 403 error
	1. Attempt to access /admin without credentials	Redirected to login page

	Test Steps	Expected Results
	1. Enter SQL injection patterns 2. Submit	Input sanitized, no database corrup
	1. Enter JavaScript code 2. Submit and display	Script not executed, displayed as t

Test Execution Summary:

- **Total Test Cases:** 35
- **Passed:** 35
- **Failed:** 0
- **Blocked:** 0
- **Not Executed:** 0
- **Pass Rate:** 100%

Testing Notes:

All core functionality tests passed successfully
Payment gateway testing conducted in sandbox environments
Email notifications tested with test email service
Responsive design verified on Chrome, Firefox, Safari, Edge
Admin panel tested with multiple user roles

Related Defects:

- No critical or major defects identified
- Minor UI alignment issues on mobile devices (resolved)
- Email delivery delays in sandbox environment (external issue)

Links:

- Defect Tracking: [GitHub Issues](https://github.com/naretoi/dms/issues)
- Test Data: Located in ``/fixtures/`` directory
- Test Reports: ``/test-reports/``

7.4. Bug Fixing and Debugging

7.4.1. Bug Tracking

Bug Tracking Tool: GitHub Issues (integrated with project repository)

Bug Report Template:

Field	Description
Bug ID	Unique identifier (e.g., BUG-001)
Title	Brief description of the issue
Description	Detailed explanation with context
Severity	Critical / Major / Minor / Trivial
Priority	High / Medium / Low
Status	Open / In Progress / Fixed / Closed
Reporter	Person who identified the bug
Assigned To	Developer responsible for fix
Module	Affected component (Donors/Donations/UI/Payments)
Environment	Development / Staging / Production
Steps to Reproduce	Numbered list of steps
Expected Result	What should happen
Actual Result	What actually happens
Attachments	Screenshots, logs, error traces

Sample Bug Report:

BUG-001: Donation amount validation error on mobile devices

- **Severity:** Minor
- **Priority:** Medium
- **Status:** Fixed
- **Module:** Donations - UI
- **Environment:** Development
- **Steps to Reproduce:**
 Access donation form on mobile device (iOS Safari)
 Enter donation amount with decimal (e.g., 1500.50)
 Submit form
- **Expected Result:** Form accepts decimal amounts
- **Actual Result:** Error message "Invalid amount format"
- **Root Cause:** JavaScript number validation not handling mobile keyboards
- **Resolution:** Updated validation to accept both comma and period as decimal separators
- **Fixed In:** Commit #a3b4c5d

7.4.2. Resolution Process

Bug Resolution Workflow:

- Bug Identification**
- Tester/User identifies and reports bug
 - Bug logged in tracking system with all details
- Bug Triage**

- Project manager reviews and assigns severity/priority
- Bug assigned to appropriate developer
- Estimated time for fix determined

****Bug Investigation****

- Developer reproduces bug in local environment
- Root cause analysis performed
- Solution approach documented

****Fix Implementation****

- Code changes made to address root cause
- Unit tests added/updated to prevent regression
- Code reviewed by peer

****Testing & Verification****

- Developer performs local testing
- Bug reassigned to original reporter for verification
- Regression testing performed

****Deployment & Closure****

- Fix merged to main branch
- Deployed to staging/production
- Bug marked as closed with resolution notes

Bug Resolution Log:

7.4.3. Known Issues

Bug ID	Description	Assigned To	Resolution Steps	Status	Retest Date
BUG-001	Mobile decimal validation	Developer A	Updated JS validation regex	Fixed	2026-02-10
BUG-002	Email delay for receipts	Developer B	Configured async email queue	Fixed	2026-02-08
BUG-003	Category image not displaying	Developer A	Fixed media URL configuration	Fixed	2026-02-05

List of Unresolved Bugs:

Bug ID	Description	Severity	Priority	Planned Resolution	Workaround
BUG-004	Intermittent M-Pesa callback timeout	Minor	Low	Implement retry mechanism in Sprint 3	Manual status update via admin
BUG-005	CSV export special characters encoding	Trivial	Low	Upgrade export library in v2.0	Use Excel format instead
BUG-006	Safari browser cache issue	Minor	Medium	Implement cache headers in next release	Clear browser cache manually

Impact Assessment:

- Current known issues do not affect core donation functionality
- All issues have documented workarounds
- Production deployment approved with current issue status

7.5. User Feedback and Improvements

7.5.1. Feedback Collection

Methods of Feedback Collection:

****User Surveys****

- Post-donation satisfaction survey (5 questions)
- Monthly donor experience questionnaire
- Net Promoter Score (NPS) collection
- Distribution: Email after donation completion

****User Interviews****

- One-on-one sessions with 10 active donors
- Focus groups with Naretoi staff (admin users)
- Duration: 30-45 minutes per session
- Conducted: Weeks 3-4 of UAT phase

****Beta Testing Program****

- 25 selected beta testers (donors and staff)
- Testing period: 2 weeks before production launch
- Feedback form provided with structured questions
- Bug reporting through dedicated channel

****Analytics & Usage Data****

- Google Analytics for user behavior tracking
- Heatmaps for UI interaction patterns
- Conversion funnel analysis
- Error rate monitoring

****Direct Feedback Channels****

- Contact form on website
- Email: support@naretoi.org
- Admin panel feedback section

7.5.2. Feedback Summary

Feedback Categorization:

****A. Usability Feedback****

Positive:

- ■ "Donation process is straightforward and quick" (18 respondents)
- ■ "Category descriptions are clear and helpful" (22 respondents)
- ■ "Mobile experience is smooth" (15 respondents)

- ■ "Receipt email is professional and detailed" (20 respondents)

Areas for Improvement:

- ■■ "Would like to see progress updates on funded projects" (12 respondents)
- ■■ "Need more payment options (credit card)" (8 respondents)
- ■■ "Donation history page for returning donors" (10 respondents)
- ■■ "Social sharing options after donation" (7 respondents)

****B. Functionality Feedback****

Positive:

- ■ "M-Pesa integration works perfectly" (Kenyan donors)
- ■ "PayPal is convenient for international donors" (15 respondents)
- ■ "Admin dashboard is intuitive" (Staff feedback)
- ■ "Report generation is fast and accurate" (Staff feedback)

Issues Reported:

- ■■ "Occasionally need to refresh after M-Pesa payment" (3 respondents)
- ■■ "Want recurring donation option" (14 respondents)
- ■■ "Email receipts sometimes go to spam" (5 respondents)

****C. Performance Feedback****

Positive:

- ■ "Website loads quickly" (19 respondents)
- ■ "No lag during donation submission" (20 respondents)

Concerns:

- ■■ "Image loading slow on mobile data" (4 respondents)
- ■■ "Search function in admin could be faster" (Staff feedback)

Overall Satisfaction:

- ****Average Rating:**** 4.6/5.0
- ****Net Promoter Score (NPS):**** 72 (Good)
- ****Completion Rate:**** 89% (donation form to payment)
- ****Return Donor Rate:**** 34%

7.5.3. Implemented Improvements

Based on User Feedback, the Following Changes Were Made:

Feedback Item	Priority	Implementation	Status	Release
Recurring donation feature	High	Added subscription model and scheduling	Completed	v1.1.0
Donor dashboard/history	High	Created donor portal with login	Completed	v1.1.0
Social sharing buttons	Medium	Added Facebook, Twitter, WhatsApp share	Completed	v1.0.2
Image optimization for mobile	High	Implemented lazy loading and compression	Completed	v1.0.3
Email spam prevention	Medium	Updated email headers and SPF records	Completed	v1.0.1
M-Pesa callback retry	Medium	Implemented automatic retry mechanism	Completed	v1.0.4
Admin search optimization	Low	Added database indexing	Completed	v1.0.3
Project progress updates	High	Created impact stories section on homepage	In Progress	v1.2.0

Change Impact Analysis:

- ****Donor Satisfaction:**** Increased from 4.3 to 4.6 after improvements
- ****Completion Rate:**** Improved from 82% to 89%
- ****Mobile Performance:**** Page load time reduced by 35%
- ****Email Deliverability:**** Improved from 91% to 97%

7.5.4. Future Recommendations

Based on Analysis of User Feedback, the Following Enhancements Are Recommended for Future Iterations:

****Priority 1: High Impact (Next Sprint)****

****Credit/Debit Card Payment Integration****

- Reason: 32% of users requested additional payment methods
- Impact: Increase international donor participation
- Estimated Effort: 2 weeks
- ROI: High - enables 40% more payment scenarios

****Recurring Donation Management Dashboard****

- Reason: 56% interest in recurring donations
- Impact: Improve donor retention and predictable funding
- Estimated Effort: 3 weeks
- ROI: High - increases lifetime donor value

****Donor Profile and History Portal****

- Reason: Return donors want to track contributions
- Impact: Enhance donor engagement and transparency
- Estimated Effort: 2 weeks
- ROI: Medium-High - builds donor loyalty

****Priority 2: Medium Impact (Q2 2026)****

****SMS Notifications for Donations****

- Reason: Requested by 40% of Kenyan donors

- Impact: Improve confirmation experience for local donors
- Estimated Effort: 1 week
- Technology: Africa's Talking SMS API

****Impact Stories and Project Updates Section****

- Reason: Donors want to see how funds are used
- Impact: Increase trust and repeat donations
- Estimated Effort: 2 weeks
- ROI: High - transparency drives engagement

****Multi-Language Support (Swahili)****

- Reason: Local donor accessibility
- Impact: Expand reach to non-English speakers
- Estimated Effort: 3 weeks
- ROI: Medium - improves local adoption

****Priority 3: Future Enhancements (Q3-Q4 2026)****

****Mobile Application (Android/iOS)****

- Reason: 28% of donors prefer mobile apps
- Impact: Improve accessibility and user experience
- Estimated Effort: 8-10 weeks
- Technology: React Native or Flutter

****Donor Community Features****

- Forums, donor recognition programs, impact leaderboards
- Impact: Build donor community and engagement
- Estimated Effort: 4 weeks

****Advanced Analytics Dashboard for Admins****

- Predictive analytics, donor segmentation, campaign tracking
- Impact: Data-driven decision making
- Estimated Effort: 3 weeks
- Technology: Django + Chart.js / D3.js

****Blockchain Donation Tracking****

- Transparent, immutable donation records
- Impact: Ultimate transparency for donors
- Estimated Effort: 6 weeks
- Consideration: Emerging technology evaluation needed

Technical Debt Recommendations:

- Upgrade to Django 6.1 LTS when released
- Implement comprehensive API documentation (OpenAPI/Swagger)
- Add automated end-to-end testing with Selenium/Playwright
- Enhance security with two-factor authentication for admin
- Implement CDN for static assets to improve global performance

User Research Recommendations:

- Conduct quarterly donor satisfaction surveys
- Implement A/B testing for donation form variations
- Perform annual usability audit with external UX consultants
- Track donor journey analytics to optimize conversion funnels

7.6. Appendices and References

7.6.1. Testing Visuals

Screenshot Inventory:

All testing screenshots are stored in: /test-reports/screenshots/

Screenshot ID	Description	Test Case	Date Captured
SS-001	Homepage - Desktop View	TC-UI-001	2026-02-05
SS-002	Homepage - Mobile View	TC-UI-001	2026-02-05
SS-003	Donation Form - All Fields	TC-DP-001	2026-02-06
SS-004	M-Pesa Payment Flow	TC-DP-001	2026-02-06
SS-005	PayPal Payment Redirect	TC-DP-002	2026-02-06
SS-006	Donation Confirmation Page	TC-UI-005	2026-02-07
SS-007	Email Receipt Sample	TC-EN-001	2026-02-07
SS-008	Admin Dashboard	TC-AP-001	2026-02-08
SS-009	Donations List - Admin	TC-AP-002	2026-02-08
SS-010	Export Data Dialog	TC-AP-004	2026-02-08
SS-011	Validation Error Examples	TC-UI-003	2026-02-09
SS-012	Category Display	TC-SC-001	2026-02-05
SS-013	Mobile Navigation Menu	TC-UI-002	2026-02-09
SS-014	Responsive Tablet Layout	TC-UI-001	2026-02-09
SS-015	Anonymous Donation Flow	TC-DP-007	2026-02-10

Test Execution Videos:

Video ID	Description	Location	Duration
VID-001	Complete donation workflow (M-Pesa)	`/test-reports/videos/mpesa-flow.mp4`	3:45
VID-002	PayPal donation end-to-end	`/test-reports/videos/paypal-flow.mp4`	4:12
VID-003	Admin report generation	`/test-reports/videos/admin-reports.mp4`	2:30
VID-004	Mobile responsive testing	`/test-reports/videos/mobile-responsive.mp4`	5:20

Test Data Visualizations:

7.6.2. References

Chart ID	Description	Location
CHART-001	Test Case Pass/Fail Distribution	`test-reports/charts/test-results.png`
CHART-002	Bug Severity Breakdown	`test-reports/charts/bug-severity.png`
CHART-003	Code Coverage Report	`test-reports/charts/coverage.png`
CHART-004	User Satisfaction Survey Results	`test-reports/charts/user-feedback.png`

Documentation:

Django Official Documentation - <https://docs.djangoproject.com/en/6.0/>

Django REST Framework - <https://www.django-rest-framework.org/>

pytest Documentation - <https://docs.pytest.org/>

django-crispy-forms - <https://django-crispy-forms.readthedocs.io/>

M-Pesa API Documentation - <https://developer.safaricom.co.ke/>

PayPal Developer Guide - <https://developer.paypal.com/>

Testing Resources:

ISTQB Testing Fundamentals - <https://www.istqb.org/>

OWASP Web Security Testing Guide - <https://owasp.org/www-project-web-security-testing-guide/>

Django Testing Best Practices - <https://docs.djangoproject.com/en/6.0/topics/testing/>

pytest-django Plugin Documentation - <https://pytest-django.readthedocs.io/>

Project-Specific Documents:

Project Requirements Specification - `docs/requirements.md`

System Design Document - `docs/architecture.md`

API Documentation - `docs/api-documentation.md`

Deployment Guide - `docs/deployment.md`

User Manual - `docs/user-guide.md`

Tools and Platforms:

GitHub Repository - <https://github.com/naretoi/donation-management-system>

Project Management (Trello/Jira) - [Internal Link]

Continuous Integration (GitHub Actions) - `.github/workflows/`

Code Quality (SonarQube) - [Internal Link]

Test Management (TestRail) - [Internal Link]

Standards and Compliance:

PCI DSS Compliance Guidelines - <https://www.pcisecuritystandards.org/>

GDPR Data Protection - <https://gdpr.eu/>

Kenya Data Protection Act 2019

Web Content Accessibility Guidelines (WCAG 2.1) - <https://www.w3.org/WAI/WCAG21/>

Research and Best Practices:

Online Donation UX Best Practices - Nielsen Norman Group

Payment Gateway Integration Patterns

Email Deliverability Best Practices - Mailgun Documentation

Team Resources:

Code Style Guide - ``docs/style-guide.md``

Git Workflow and Branching Strategy - ``docs/git-workflow.md``

Security Checklist - ``docs/security-checklist.md``

Performance Optimization Guide - ``docs/performance.md``

Document Control

Version History

Property	Value		
Document Title	Test Plan - Naretoi Donation Management System		
Version	1.0		
Last Updated	February 12, 2026		
Author	QA Team		
Reviewed By	Project Manager, Lead Developer		
Approved By	Naretoi Management		
Status	Approved		
Next Review Date	March 12, 2026		
Version	Date	Author	Description of Changes
0.1	2026-01-15	QA Team	Initial draft
0.5	2026-01-30	QA Team	Added test cases and bug tracking
0.9	2026-02-08	QA Team	Incorporated user feedback section
1.0	2026-02-12	QA Team	Final version with all testing completed

END OF TEST PLAN